

Direction I: Fill In Blank Space(3 points)

1. If the acceleration of a moving body is zero, then the kinetic energy of the body is _____
2. When a body falls from the top of a building, its kinetic energy _____ but _____ remains constant.

Direction II: Choose the correct answer from the given alternatives and put on black space only (6 points)

1. ■ An object, initially at rest, covers a distance of 10m in 4s under the action of a constant force. If the work done by the force during this time is 50J, the mass of the object is
A. 4kg B. 3kg C. 5kg D. 6kg
2. ■ A 4kg object is accelerated uniformly from rest to a speed of 16m/s in 8s. What power is delivered to the object just after 6s?
A. 112W B. 132W C. 48W D. 164W
3. ■ A car of mass 2000kg moving at a speed of 20m/s slows down uniformly to a speed of 10m/s in 4s. The work done on the car during the first 2 seconds is
A. $1.75 \times 10^5 \text{ J}$ B. $-1.75 \times 10^5 \text{ J}$ C. $3 \times 10^5 \text{ J}$ D. $-3 \times 10^5 \text{ J}$
4. ■ The momentum of a 10kg ball is equal to the momentum of a 4kg block traveling at a speed of 5m/s. What is the kinetic energy of the ball?
A. 40J B. 20J C. 4J D. 5J
5. ■ A block slides down a plane inclined at an angle of 53° above the horizontal with an acceleration of 5 m/s^2 . What is the coefficient of sliding friction between the block and the plane? ($\sin 53^\circ = 0.8$ and $\cos 53^\circ = 0.6$)
A. 0.4 B. 0.5 C. 0.3 D. 0.6
6. ■ A ball is thrown at an angle of 53° above the horizontal with an initial speed of 20m/s. How high does the ball rise
A. 12.8m B. 128m C. 64m D. None

Direction III: Work out part

Show all the necessary steps accordingly (6 points)

3. A stone was thrown vertically with a speed of 20m/s . 1s later a second stone was released from a height of 25m .

I. At what height did they meet?

II. What is the speed of the first stone be when they meet?

4. If the vector $2\mathbf{i}+2\mathbf{j}+6\mathbf{k}$ is perpendicular to the vector $2\mathbf{j}-2\mathbf{i}+\beta\mathbf{k}$, then the value of β is?

5. If $\mathbf{a} \times \mathbf{b} = \sqrt{3}\mathbf{a} \cdot \mathbf{b}$, then the value of $|\mathbf{a} + \mathbf{b}|$